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INSTITUTE OF TECHNOLOGY
DELHI CAMPUS.

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The United Nations General Assembly
(Disarmament and International Security Committee)

BACKGROUND GUIDE

Agenda:

"Establishing an International Regulatory Framework for the Development and Deployment of Lethal Autonomous Weapons Systems."

Letter from the Executive Board

Dear Delegates,

We, the executive board members of the UNGA-DISEC, find immense pleasure in being a part of the IITD MUN'26. We extend a warm welcome to all of you.

We believe that 'study guides' are detrimental to the individual growth of the members since they overlook a very important aspect of this activity, which is Research. We are sure that this background guide gives you a perfect launching pad to start your research. The Background guide would be as abstract as possible and would just give you a fundamental perspective on what the executive board believes you should know before you commence your research. This being clear, kindly do not limit your research to the areas highlighted; further, ensure that you logically deduce and push your research to areas associated with the issues mentioned. The objective of this background guide is to provide you with a 'background' of the issue at hand, and therefore it might seem to some as not being comprehensive enough. We are not looking for existing solutions or strategies that would be a copy-paste of what the countries you are representing have already stated; instead, we seek an out-of-the-box solution from you.

Wishing you all very good luck, and hope to see you all at this conference discussing important issues of international interest. We look forward to meeting you all at the MUN, and please do not hesitate to contact us if you have any questions before attending the MUN.

Looking forward to seeing you all.

Regards,

Anshika Singh - Chairperson.

Points To Remember

1. Procedure:

The purpose of putting procedural rules in any committee is to ensure a more organized and efficient debate. The committee will follow the UNA USA Rules of Procedure. Although the Executive Board shall be fairly strict with the Rules of Procedure, the discussion of the agenda will be the main priority. So, delegates are advised not to restrict their statements due to hesitation regarding the procedure.

2. Foreign Policy:

Following the foreign policy of one's country is the most important aspect of a Model UN Conference. This is what essentially differentiates a Model UN from other debating formats. To violate one's foreign policy without adequate reason is one of the worst mistakes a delegate can make.

3. Role of the Executive Board:

The Executive Board is appointed to facilitate debate. The committee shall decide the direction and flow of the debate. The delegates are the ones who constitute the committee and hence must be uninhibited while presenting their opinions/stances on any issue. However, the Executive Board may put forward questions and/or ask for clarifications at all points in time to debate further and test participants.

4. Nature of Source/Evidence:

This Background Guide is meant solely for research purposes and must not be cited as evidence to substantiate statements made during the conference. Evidence or proof for substantiating statements made during the formal debate is acceptable from the following sources:

United Nations:

Documents and findings by the United Nations or any related UN body are held as credible proof to support a claim or argument. Multilateral Organizations:

Documents from international organizations like OIC, NAFTA, SAARC, BRICS, EU, ASEAN, the International Criminal Court, etc. may also be presented as credible sources of information

Government Reports:

These reports can be used in a similar way as the State-operated News Agencies reports and can, in all circumstances, be denied by another country.

Valid Evidence In the Committee

Evidence or proof from the following sources will be accepted as credible by the Executive Board:

- **State-operated News Agencies:**

These reports can be used in support of or against the State that owns the News Agency. These reports, if credible or substantial enough, can be used to support or against any country as such, but in that situation, they can be denied by any other country in the council. Some examples are,

1. RIA Novosti (Russia)
2. IRNA (Iran)
3. BBC & Reuters (United Kingdom)
4. Al Jazeera (Qatar)
5. Xinhua News Agency (PR China)

- **Government Reports:**

These reports can be used in a similar way as the State Operated News Agencies reports and can, in all circumstances, be denied by another country. However, a nuance is that a report that is being denied by a certain country can still be accepted by the Executive Board as credible information. Some examples are,

1. Government Websites like the State Department of the United States of America or the Ministry of Defense of the Russian Federation
2. Ministry of Foreign Affairs of various nations like India or China
3. Permanent Representatives to the United Nations Reports
4. Multilateral Organizations like NATO.

- **United Nations Reports:**

All UN Reports are considered credible information or evidence for the Executive Board of the committee

I) UN Bodies like the UNHRC, UNODC, etc.

II) UN Affiliated Bodies like UNICEF, International Committee of the Red Cross, etc.

III) Treaty Bodies like the United Nations Convention on the Rights of the Child, etc.

- ***Please note- Reports from NGOs working with UNESCO, UNICEF, and other UN bodies will be accepted. Under no circumstances will sources like Wikipedia or newspapers like the Guardian, Times of India, etc. be accepted.***

ABOUT THE COMMITTEE

Introduction

The United Nations General Assembly is one of the six principal organs of the United Nations (UN) and the only body in which every member of the organization is represented and allowed to vote. The first session of the assembly convened on Jan. 10, 1946, in London, with 51 countries represented. As of 2006, there were 192 members of the General Assembly. Numerous non-members, such as states, organizations, and other entities (e.g., the Vatican, the African Union, the International Committee of the Red Cross, and Palestine), maintain observer status, enabling them to participate in the work of the General Assembly. The General Assembly exercises deliberative, supervisory, financial, and elective functions relating to any matter within the scope of the UN Charter. Its primary role, however, is to discuss issues and make recommendations, though it has no power to enforce its resolutions or compel state action. Other functions include admitting new members; selecting members of the Economic and Social Council, the non-permanent members of the Security Council, and the Trusteeship Council; supervising the activities of the other UN organs, from which the General Assembly receives reports; and participating in the election of judges to the International Court of Justice and the selection of the secretary-general. Decisions usually are reached by a simple majority vote. On important questions, however, such as the admission of new members, budgetary matters, and peace and security issues, a two-thirds majority is required. The General Assembly convenes annually and in special sessions, electing a new president each year from among five regional groups of states. At the beginning of each regular session, the General Assembly also holds a general debate, in which all members participate and may raise any issue of international concern. Most work, however, is delegated to six main committees, known as

- (1) Disarmament and International Security
- (2) Economic and Financial
- (3) Social, Humanitarian, and Cultural
- (4) Special Political and Decolonization
- (5) Administrative and Budgetary
- (6) Legal

The large size of the General Assembly and the diversity of the issues it discussed contributed to the emergence of regionally based voting blocs in the 1960s. During the Cold War, the Soviet Union and the countries of Eastern Europe formed one of the most cohesive blocs. Since the 1980s and the end of the Cold War, blocs have formed around “North-South” economic issues, i.e., issues of disagreement between the more prosperous, industrialized countries of the Northern Hemisphere and the poorer, less-industrialized developing countries of the Southern Hemisphere. The number of resolutions passed by the General Assembly each year has climbed to more than 300, and many resolutions are adopted without opposition. Nevertheless, there have been sharp disagreements among members on several issues, such as those relating to the Cold War, the Arab-Israeli conflict, and human rights.

1. The Role and Functions of the UNGA

- a. Promoting International Cooperation and Peace
- b. Setting the Global Agenda
- c. Reviewing the Work of Specialized Agencies

The UNGA acts as a platform for member states to engage in diplomatic dialogue, fostering understanding and cooperation. It provides a space for nations to voice their concerns, propose solutions, and resolve conflicts. It also plays a significant role in setting the global agenda by addressing pressing issues facing humanity, influencing the policy priorities of the UN, and shaping the direction of international cooperation. The UNGA oversees the work of specialized agencies within the UN system. It ensures that these agencies align their activities with the broader goals and principles of the organization.

The Assembly is empowered to make recommendations to States on international issues within its competence. It has also initiated actions- political, economic, humanitarian, social and legal- which have affected the lives of millions of people throughout the world.

The landmark Millennium Declaration, adopted in 2000, and the 2005 World Summit Outcome Document, reflect the commitment of Member States:

- to reach specific goals to attain peace, security, and disarmament along with development and poverty eradication;
- to safeguard human rights and promote the rule of law; to protect our common environment;
- to meet the special needs of Africa; and to strengthen the United Nations

In September 2015, the Assembly agreed on a set of 17 Sustainable Development Goals, contained in the outcome document of the United Nations summit for the adoption of the post-2015 development agenda (resolution 70/1).

According to the Charter of the United Nations, the General Assembly may:

- Consider and approve the United Nations budget and establish the financial assessments of Member States;
- Elect the non-permanent members of the Security Council and the members of other United Nations councils and organs and, on the recommendation of the Security Council, appoint the Secretary-General
- Consider and make recommendations on the general principles of cooperation for maintaining international peace and security, including disarmament;
- Discuss any question relating to international peace and security and, except where a dispute or situation is currently being discussed by the Security Council, make recommendations on it;
- Discuss, with the same exception, and make recommendations on any questions within the scope of the Charter or affecting the powers and functions of any organ of the United Nations;
- Initiate studies and make recommendations to promote international political cooperation, the development and codification of international law, the realization of human rights and fundamental freedoms, and international collaboration in the economic, social, humanitarian, cultural, educational, and health fields;
- Make recommendations for the peaceful settlement of any situation that might impair friendly relations among countries;
- Consider reports from the Security Council and other United Nations organs.

The Assembly may also take action in cases of a threat to the peace, breach of peace, or act of aggression, when the Security Council has failed to act owing to the negative vote of a permanent member. In such instances, according to its “Uniting for Peace” resolution of 3 November 1950, the Assembly may consider the matter immediately and recommend to its Members collective measures to maintain or restore international peace and security.

2. The Composition of the UNGA

The UNGA is composed of all *193 member states* of the United Nations, making it a truly global body. Each member state, regardless of its size or population, has an *equal vote*, ensuring the principle of sovereign equality. To facilitate discussions and decision-making, member states are organized into *five regional groups*: African Group, Asian Group, Eastern European Group, Latin American and Caribbean Group, and Western European and Others Group.

3. Resolutions and Voting

A resolution is a formal expression of the UNGA's opinion or recommendation on a particular matter. Resolutions are adopted through voting, with a two-thirds majority required for important decisions on issues such as peace and security, admission of new members, and budgetary matters.

4. Committee Deliberations and Consensus Building

Much of the UNGA's work is conducted through its committees and subsidiary bodies. These committees provide a platform for in-depth discussions and negotiations on specific issues. Consensus building is often emphasized to find common ground.

The First Committee: Disarmament and International Security (DISEC)

The First Committee, formally known as the Disarmament and International Security Committee (DISEC), stands as one of the most crucial subsidiary bodies of the United Nations General Assembly. Established as part of the original six main committees, DISEC serves as the primary forum within the UN system for multilateral discussions on disarmament and international security matters.

Mandate and Scope

DISEC's mandate encompasses a broad spectrum of security-related issues, including nuclear disarmament, conventional weapons regulation, regional security arrangements, and emerging security challenges. The committee addresses both traditional security concerns and contemporary threats such as cyber warfare, autonomous weapons systems, and the militarization of outer space. Its work directly supports the UN Charter's

fundamental goal of maintaining international peace and security through collective action and diplomacy.

The committee operates under the principle that disarmament and international security are interconnected and essential for global stability. It recognizes that excessive military expenditure, arms races, and the proliferation of weapons of mass destruction pose significant threats to international peace and development. Through its deliberations, DISEC seeks to promote transparency, build confidence among nations, and establish normative frameworks for responsible state behavior in security matters.

Key Areas of Focus

Nuclear disarmament remains one of DISEC's primary concerns, with the committee regularly addressing issues such as the Non-Proliferation Treaty (NPT), nuclear testing, and the establishment of nuclear-weapon-free zones. The committee has been instrumental in promoting dialogue on reducing nuclear arsenals and preventing the spread of nuclear weapons technology. It also addresses the humanitarian consequences of nuclear weapons and supports initiatives for their complete elimination.

Conventional weapons control constitutes another significant area of the committee's work. This includes discussions on the Arms Trade Treaty, small arms and light weapons proliferation, landmines, cluster munitions, and emerging weapons technologies. DISEC recognizes that conventional weapons, while legally possessed by states, can fuel conflicts and instability when traded irresponsibly or accumulated excessively.

Regional security issues also feature prominently in DISEC's agenda. The committee provides a platform for member states to address specific regional tensions, confidence-building measures, and cooperative security arrangements. It acknowledges that regional approaches to security can complement global efforts and contribute to overall international stability.

Working Methods and Procedures

DISEC operates through a combination of formal meetings, thematic debates, and interactive dialogues. The committee typically convenes during the General Assembly's regular session, usually from October to November, bringing together all 193 UN member states. Each delegation has equal representation and voting rights, reflecting the democratic principles of the General Assembly.

The committee's work is organized around specific agenda items, with member states presenting position papers, engaging in debates, and negotiating draft resolutions. These resolutions, once adopted by the committee, are forwarded to the General Assembly plenary for final consideration and adoption. While DISEC resolutions are not legally

binding, they carry significant political and moral weight, influencing international norms and state behavior.

The committee also regularly invites expert briefings from international organizations, research institutions, and civil society groups to inform its deliberations. This inclusive approach ensures that discussions benefit from diverse perspectives and technical expertise, enhancing the quality and relevance of the committee's work.

ABOUT THE AGENDA

Introduction

The intersection of artificial intelligence and military technology has precipitated a profound paradigm shift in the fundamental nature of warfare. Historically, the application of lethal force has been inherently linked to human agency, human deliberation, and human responsibility. The emergence of Lethal Autonomous Weapons Systems fundamentally challenges this historical precedent. These systems possess the capability to identify, track, and engage targets without subsequent human intervention once activated. As the Disarmament and International Security Committee convenes, the primary objective is to address the multifaceted challenges posed by the militarization of artificial intelligence. The rapid pace of technological innovation has drastically outstripped the evolution of global diplomatic and legal norms. Consequently, the international community faces an urgent imperative to formulate a cohesive regulatory architecture that governs how these technologies are designed, tested, and utilized on the battlefield.

The integration of autonomy into weapons systems is not a sudden phenomenon but rather the culmination of decades of incremental advancements in robotics, sensor technology, and machine learning algorithms. However, the transition from automated defensive systems to offensive platforms capable of independent lethal decision-making represents a critical threshold. Crossing this threshold raises profound questions about human dignity, the laws of armed conflict, and the stability of international security. Without a universal regulatory mechanism, states are left to rely on ambiguous interpretations of existing international law, leading to a fragmented and potentially dangerous global security environment. The responsibility falls upon the international community to bridge the gap between technological capability and legal accountability.

The urgency of this issue is compounded by the dual-use nature of the underlying technologies. The same artificial intelligence advancements that drive autonomous vehicles, medical diagnostics, and civilian logistics can be rapidly repurposed for military applications. This democratization of advanced technology means that the capacity to develop Lethal Autonomous Weapons Systems is not restricted to a small number of heavily industrialized nations. Non-state actors and nations with limited military budgets can potentially acquire and deploy sophisticated autonomous systems. The proliferation risks associated with these weapons are unprecedented, demanding a regulatory framework that is robust, adaptable, and universally applicable.

Furthermore, the introduction of autonomous systems introduces severe risks of unintended escalation and strategic instability. When machines operate at speeds beyond human comprehension, the interaction between competing autonomous systems could trigger a rapid escalation of hostilities, commonly referred to as a flash war. The lack of predictability inherent in complex machine learning algorithms means that even the developers of these systems cannot guarantee their behavior in novel, unstructured environments. Therefore, the establishment of an international regulatory framework is not merely a legal exercise but an essential endeavor to preserve global peace, maintain human control over the use of force, and prevent the dawn of an uncontrollable algorithmic arms race.

Background of the Agenda

The contemporary geopolitical landscape is defined by a fierce competition for technological supremacy, with artificial intelligence recognized as the most critical domain of this rivalry. Major military powers have openly declared that leadership in artificial intelligence is synonymous with future economic and military dominance. Within this context, the development of Lethal Autonomous Weapons Systems is heavily driven by the desire to achieve a tactical and strategic overmatch against potential adversaries. Military strategists argue that autonomous systems offer significant advantages on the modern battlefield. These systems can process vast amounts of data from diverse sensor networks simultaneously, react to threats at machine speed, operate in environments where communication links are jammed or severed, and undertake high-risk missions without jeopardizing human personnel.

Despite these perceived military advantages, the integration of lethal autonomy creates severe diplomatic and ethical fissures within the international community. A significant portion of the global populace, alongside numerous non-governmental organizations and technological experts, argues that delegating the decision to take human life to a machine is inherently immoral. This perspective asserts that human dignity requires that lethal decisions be made by moral agents who can comprehend the value of human life and the gravity of destroying it. Algorithms, devoid of empathy, compassion, and moral reasoning, cannot fulfill this requirement. This ethical revulsion has fueled a prominent global campaign calling for a preemptive ban on fully autonomous weapons, mirroring past successful efforts to ban biological weapons, chemical weapons, and anti-personnel landmines.

The debate is further complicated by the lack of consensus on the technical definitions and thresholds of autonomy. States possess divergent views on what constitutes "meaningful human control" and at what specific point a weapon transitions from being a highly automated tool to a fully autonomous entity. Some nations advocate for a binding legal instrument that explicitly prohibits systems lacking human control. Conversely, other technologically advanced nations argue that existing international humanitarian law is entirely sufficient to regulate these new weapons, provided states rigorously apply established principles such as distinction and proportionality. These states often express concern that a premature or overly broad ban could stifle legitimate technological research and deprive them of systems that could potentially reduce civilian casualties by executing strikes with unprecedented precision.

The current diplomatic stalemate is a direct consequence of these opposing strategic interests and ethical frameworks. The Disarmament and International Security Committee must navigate this complex web of national security priorities, humanitarian concerns, and technological realities. The background of this agenda is deeply rooted in the broader crisis of global arms control. Traditional arms control treaties were designed for physical, easily verifiable assets like nuclear warheads, ballistic missiles, and troop formations. Regulating algorithms, software updates, and lines of code requires an entirely novel approach to verification, compliance, and transparency. As such, addressing this agenda requires not only legal innovation but also a fundamental reimagining of how the international community monitors and controls the proliferation of intangible, software-driven military capabilities.

Agenda Breakdown

To thoroughly comprehend the scope of the committee mandate, it is essential to deconstruct the agenda title into its constituent elements. Analyzing each keyword independently provides clarity on the specific actions, targets, and goals that the committee is expected to address.

Establishing: This active verb dictates the primary objective of the committee. It implies that the current status quo is insufficient and that the committee is tasked with creating, building, or formalizing a new structure. Establishing goes beyond mere discussion or observation. It requires the formulation of tangible, actionable mechanisms. This could manifest as drafting the foundational principles for a new treaty, creating voluntary guidelines, setting up technical standards, or forming a new international oversight body. The focus is on creation and implementation.

International Regulatory Framework: This phrase outlines the desired outcome of the establishment process. It is a broad term that encompasses various legal and diplomatic instruments. It does not strictly demand a legally binding treaty, although that is one possible interpretation. A regulatory framework consists of the rules, norms, protocols, and verification mechanisms that govern state behavior. It could include a combination of hard law (binding treaties and conventions) and soft law (political declarations, codes of conduct, and technical standards). The framework must be international, meaning it requires broad multilateral consensus and applicability across different jurisdictions and geopolitical blocs.

Development: This term targets the entire lifecycle of a weapon system before it ever reaches a battlefield. Development encompasses theoretical research, algorithmic design, machine learning training processes, physical manufacturing, and rigorous testing phases. Regulating development is extraordinarily difficult because much of the foundational research in artificial intelligence is dual-use and occurs in civilian academic or corporate settings. Regulating development requires states to implement internal oversight mechanisms to ensure that domestic research does not cross the threshold into prohibited lethal autonomy.

Deployment: This refers to the actual fielding, operationalization, and use of these systems in a combat environment or theater of war. Deployment is the stage where the weapon is activated and capable of executing its programmed functions. Regulating deployment involves establishing rules of engagement, defining the geographical and temporal limits of autonomous operations, and ensuring that there are mechanisms for human intervention or fail-safes in case the system malfunctions or encounters unanticipated scenarios.

Lethal Autonomous Weapons Systems: This is the subject of the regulation. "Lethal" restricts the scope to systems designed or capable of killing or causing severe physical destruction. It generally excludes non-lethal autonomous systems used for logistics, medical evacuation, or purely intelligence gathering. "Autonomous" is the most contested word. In this context, it refers to a system capable of selecting and engaging targets without human intervention after initial activation. "Weapons Systems" refers to the integration of the lethal payload, the sensors, and the processing unit into a cohesive platform.

The entire agenda, "Establishing an International Regulatory Framework for the Development and Deployment of Lethal Autonomous Weapons Systems," charges the committee with a monumental task. The committee must design a comprehensive global architecture (framework) that dictates the legal and ethical boundaries (regulatory) of how nations invent and build (development) as well as how they field and use (deployment) machines that can independently decide to kill (lethal autonomous weapons systems). The agenda requires bridging the gap between those who seek an outright ban and those who believe existing laws are sufficient, ultimately finding a consensus mechanism that preserves human control over the use of force while acknowledging the unstoppable trajectory of technological advancement.

Keywords

Meaningful Human Control: This is arguably the most critical concept in the discourse surrounding autonomous weapons. It refers to the principle that humans must retain a significant and decisive role in the use of lethal force. While there is no single universally agreed-upon definition, it generally requires that a human operator possesses sufficient contextual information about the target, the weapon system, and the environment to make an informed and deliberate decision before a lethal strike occurs. It also implies that the human has the technical ability and the time to abort an attack if the situation changes. The debate centers on exactly how much control is "meaningful" and at what stage of the targeting cycle that control must be exercised.

Artificial Intelligence: In the context of this agenda, artificial intelligence refers to computer systems capable of performing tasks that typically require human intelligence. This includes visual perception, speech recognition, decision-making, and translation between languages. For weapons systems, artificial intelligence allows the platform to analyze complex sensor data, recognize patterns, identify specific target signatures, and navigate dynamic environments without human guidance.

Machine Learning: A specific subset of artificial intelligence where algorithms are trained on vast datasets to recognize patterns and make predictions or decisions without being explicitly programmed for every specific scenario. Machine learning is crucial for autonomous weapons because it allows them to adapt to new environments and identify targets based on learned characteristics rather than strict, pre-coded rules. The challenge is that machine learning models can be unpredictable, and their decision-making processes are often opaque.

Algorithmic Bias: This refers to systematic and repeatable errors in a computer system that create unfair outcomes. In the context of lethal weapons, algorithmic bias is a severe danger. If the datasets used to train a targeting algorithm primarily feature specific demographics, clothing styles, or behavioral patterns, the system may disproportionately misidentify civilians from certain regions or ethnic groups as combatants. This technical flaw directly violates the principle of distinction in international humanitarian law.

Distinction: A foundational pillar of International Humanitarian Law. The principle of distinction requires that parties to a conflict must at all times distinguish between the civilian population and combatants, and between civilian objects and military objectives. Attacks may only be directed against military objectives. The core technological challenge for autonomous weapons is whether an algorithm can accurately make this highly context-dependent distinction in chaotic, urban combat environments.

Proportionality: Another cornerstone of International Humanitarian Law. The principle of proportionality prohibits an attack which may be expected to cause incidental loss of civilian life, injury to civilians, damage to civilian objects, or a combination thereof, which would be excessive in relation to the concrete and direct military advantage anticipated. Assessing proportionality requires nuanced human judgment, moral reasoning, and the ability to weigh disparate values (civilian harm versus military gain), a calculation that currently exceeds the capabilities of any artificial intelligence.

Accountability Gap: This term describes the legal and moral void created when an autonomous weapon commits a war crime or causes unintended civilian casualties. Traditional law holds individuals (soldiers, commanders) accountable for their actions. If an algorithm independently decides to strike a civilian target, determining who is legally responsible is highly complex. Is it the programmer, the manufacturer, the commander who deployed the system, or the state itself? The accountability gap threatens to undermine the entire enforcement mechanism of the laws of armed conflict.

Dual Use Technology: This refers to technology that has both civilian and military applications. The vast majority of the hardware (processors, sensors) and software (neural networks, computer vision) required for autonomous weapons is developed in the commercial sector for self-driving cars, facial recognition, and automated logistics. This dual-use nature makes it nearly impossible to control the proliferation of the underlying technology through traditional export controls or nonproliferation regimes.

Swarming: A tactical concept where multiple, relatively inexpensive autonomous systems (often drones) operate collaboratively as a single, cohesive network to overwhelm an adversary's defenses. In a swarm, individual units communicate with each other, share sensor data, and coordinate their actions without central human direction. Swarming drastically increases the complexity of the battlefield and the speed of engagement, heightening the risks of escalation and loss of human control.

Black Box Problem: This refers to the opacity of advanced machine learning models, particularly deep neural networks. Even the engineers who design these systems often cannot explain exactly how or why the algorithm arrived at a specific decision. In a military context, the black box problem means that if an autonomous weapon makes a targeting error, investigators may be unable to trace the logical steps that led to the failure. This lack of explainability directly impedes legal reviews, accountability, and the ability to trust the system in future operations.

Historical Context and Developments

The technological lineage of Lethal Autonomous Weapons Systems extends back several decades, rooted in the gradual automation of defensive military platforms. During the Cold War, the increasing speed and volume of aerial and missile threats necessitated defensive systems that could react faster than human operators. The introduction of the Phalanx Close-In Weapon System by the United States Navy in the late 1970s marked a significant milestone. The Phalanx is an automated gun system designed to defend ships against incoming anti-ship missiles. Once placed in its fully automatic mode, the system uses its own radar to detect, track, evaluate, and engage incoming threats without human intervention. Similarly, the Aegis Combat System, also developed in the late twentieth century, possesses automated modes to protect naval task forces from massive, simultaneous aerial attacks. These early systems established the precedent that, in specific defensive contexts where human reaction time is insufficient, delegating lethal engagement to a machine is militarily necessary and legally acceptable.

The subsequent major development was the proliferation of Unmanned Aerial Vehicles, commonly known as drones, primarily in the years following the terrorist attacks of September 2001. While early combat drones like the MQ-1 Predator and MQ-9 Reaper were strictly remotely piloted by human operators, they represented a crucial step in decoupling the human from the physical location of the weapon. These systems accustomed militaries and the global public to the concept of robotic warfare. Over time, these platforms became increasingly automated in their flight, navigation, and surveillance capabilities. The human operator was reduced from physically flying the aircraft to monitoring its systems and occasionally authorizing a strike. This shift from "human in the loop" (direct control) to "human on the loop" (supervisory control) laid the groundwork for complete autonomy.

Simultaneously, the development of loitering munitions, often referred to as kamikaze drones, further blurred the lines between guided missiles and autonomous aircraft. Systems like the Israeli-designed Harpy, introduced in the late 1980s and 1990s, were designed to be launched into a designated area, independently loiter in the sky, detect enemy radar emissions, and

automatically dive into the target to destroy it. While the parameters were narrowly defined and the target type was specific (radar installations), the weapon executed the final targeting and engagement sequence independently. This represented a shift from purely defensive automation (like the Phalanx) to offensive, autonomous target selection within a specific geographic box.

The true inflection point in the historical context occurred in the mid 2010s, driven by massive breakthroughs in commercial artificial intelligence, specifically the advent of deep learning and advanced computer vision. Suddenly, algorithms were capable of identifying specific objects, faces, and behaviors in complex, cluttered environments with a high degree of accuracy. Military research and development rapidly pivoted to integrate these commercial advancements into weapons platforms. The goal shifted from simply hitting a programmed coordinate or a distinct radar emission to deploying systems that could visually identify enemy combatants or specific vehicle types in dynamic environments and engage them independently.

Diplomatically, the international community began to recognize the gravity of this technological shift around 2013. In that year, Christof Heyns, the United Nations Special Rapporteur on extrajudicial, summary or arbitrary executions, presented a seminal report to the Human Rights Council. His report warned of the profound ethical and legal challenges posed by lethal autonomous robotics and called for national moratoria on their development. This report acted as a catalyst, sparking intense global debate. Consequently, the issue was brought before the Convention on Certain Conventional Weapons (CCW) in Geneva. In 2014, the CCW held its first informal meeting of experts on the subject. By 2017, acknowledging the urgency of the issue, the CCW established a formal Group of Governmental Experts on Lethal Autonomous Weapons Systems.

Over the past several years, this Group of Governmental Experts has been the primary international forum for discussing this agenda. The technological reality has continued to outpace the diplomatic process. While diplomats debated definitions in Geneva, militaries around the world accelerated their testing and integration of artificial intelligence in targeting systems. The conflicts of the 2020s, particularly in the Caucasus, the Middle East, and Eastern Europe, have witnessed the unprecedented use of highly automated and semi-autonomous systems, demonstrating that algorithmic warfare has moved from theoretical research to active deployment. The historical trajectory shows a clear, accelerating trend from human-operated tools to automated defensive systems, and now toward offensive platforms possessing artificial intelligence capabilities.

Detailed Analysis of Problem

The deployment of Lethal Autonomous Weapons Systems generates a multidimensional crisis that challenges the foundations of international law, moral philosophy, and global security strategy. To establish an effective regulatory framework, it is vital to analyze these problems in granular detail.

The most profound dilemma is ethical, centering on the concept of human dignity. The decision to terminate a human life is the most severe action an individual or a state can take. Philosophers and legal scholars argue that this decision must be accompanied by moral agency, compassion, and the capacity for ethical reasoning. An algorithm, regardless of its sophistication, is a mathematical construct operating on statistical probabilities. It does not understand the value of the life it is extinguishing, nor does it comprehend the suffering caused by its actions. Delegating the authority to kill to an unfeeling machine is viewed by many as a fundamental degradation of human dignity. It reduces human beings to mere data points and pixel configurations to be processed and eliminated by a software program. This ethical objection forms the core of the argument for a total, preemptive prohibition of fully autonomous weapons.

From a legal perspective, the primary problem is ensuring compliance with International Humanitarian Law, specifically the principles of distinction and proportionality. In a conventional war zone, combatants and civilians are often intermingled. A human soldier uses contextual clues to distinguish a civilian carrying a farm implement from a combatant carrying a weapon. Furthermore, a human soldier assessing proportionality must weigh the military necessity of a strike against the potential for civilian harm. This involves subjective moral judgment. For example, calculating whether destroying a sniper position justifies the potential destruction of an adjacent civilian dwelling. Current artificial intelligence lacks the contextual understanding, common sense, and moral reasoning required to make these highly nuanced determinations. If an autonomous weapon cannot reliably adhere to these principles in complex environments, its deployment is inherently unlawful under the laws of armed conflict.

Compounding the legal problem is the aforementioned accountability gap. International criminal law is built upon the premise of individual responsibility. If a soldier commits a war crime, they are prosecuted. If a commander orders a war crime, or fails to prevent one they knew about, they are held liable under the doctrine of command responsibility. However, if a fully autonomous system, operating without direct human control, unexpectedly targets a civilian hospital due to a glitch, a novel environmental variable, or algorithmic bias, assigning criminal liability becomes deeply problematic. The machine cannot be punished. The programmer may have written the code years prior without malicious intent. The commander who deployed the weapon may have reasonably believed it would function correctly based on its testing data. If no individual can be held legally responsible for war crimes committed by autonomous weapons, the deterrent effect of international law evaporates, potentially leading to increased atrocities and impunity on the battlefield.

Strategically and technically, the integration of autonomous weapons introduces the severe risk of uncontrolled escalation and conflict instability. Artificial intelligence operates at speeds incomprehensible to the human mind. If two adversary nations deploy autonomous swarms in proximity, these systems will interact, react, and counter-react at machine speed. A minor incident, a sensor glitch, or a misinterpretation of an adversary's algorithmic behavior could trigger a rapid, escalating spiral of lethal force. This phenomenon, often termed a flash war, could result in a massive military engagement before human political or military leaders even realize a

conflict has begun, completely stripping them of the ability to manage the crisis or negotiate a de-escalation.

Furthermore, artificial intelligence systems, particularly those relying on machine learning, are highly vulnerable to novel forms of attack. Adversaries can utilize adversarial machine learning techniques to subtly alter the physical environment or manipulate sensor data to deceive the autonomous system. A slightly modified stop sign might be interpreted by an algorithm as a speed limit sign, or in a military context, a civilian bus might be mathematically manipulated to appear as a tank to the weapon's optical sensors. These vulnerabilities, combined with the "black box" nature of neural networks, mean that autonomous weapons can fail in unpredictable, catastrophic, and deeply unintuitive ways, making them unreliable and highly dangerous tools for national security.

International Legal Framework

The foundational challenge in regulating Lethal Autonomous Weapons Systems is that international law currently lacks an explicit, universally ratified treaty that governs them by name. Consequently, legal analysis relies on interpreting the existing corpus of International Humanitarian Law and determining how its provisions apply to algorithmic entities. Any regulatory framework established by this committee must be anchored in, or built upon, these foundational texts.

The primary legal text governing the conduct of hostilities is the Geneva Conventions of 1949 and, most importantly in this context, the Additional Protocol I of 1977. Additional Protocol I establishes the core rules for the protection of victims of international armed conflicts and sets limits on the methods and means of warfare.

Article 36 of Additional Protocol I is the cornerstone of the legal debate surrounding the development of new weapons. It mandates that "In the study, development, acquisition or adoption of a new weapon, means or method of warfare, a High Contracting Party is under an obligation to determine whether its employment would, in some or all circumstances, be prohibited by this Protocol or by any other rule of international law applicable to the High Contracting Party." This requires states to conduct rigorous legal reviews of autonomous weapons before they are fielded. The difficulty arises from the nature of machine learning. If a weapon system is designed to continuously learn and alter its targeting parameters after deployment based on new data, a static legal review conducted during the development phase becomes obsolete the moment the weapon enters the battlefield. This dynamic nature of artificial intelligence severely complicates state compliance with Article 36 obligations.

Article 48 of Additional Protocol I codifies the Basic Rule of distinction. It states that "the Parties to the conflict shall at all times distinguish between the civilian population and combatants and between civilian objects and military objectives and accordingly shall direct their operations only against military objectives." As analyzed previously, the legal burden rests on whether the sensor suites and processing algorithms of an autonomous weapon can meet this stringent requirement in

dynamic, unstructured environments, such as urban warfare, where combatants often do not wear distinct uniforms.

Article 51(4) and 51(5) of Additional Protocol I elaborate on the prohibition of indiscriminate attacks and the principle of proportionality. Article 51(5)(b) specifically prohibits attacks "which may be expected to cause incidental loss of civilian life, injury to civilians, damage to civilian objects, or a combination thereof, which would be excessive in relation to the concrete and direct military advantage anticipated." This requires a forward-looking, value-based judgment comparing human suffering against military gain. Critics argue that transferring this heavily subjective legal obligation to a mathematical algorithm is a fundamental violation of the Protocol.

Article 57 of Additional Protocol I dictates the precautions that must be taken in attack. It requires those who plan or decide upon an attack to do everything feasible to verify that the objectives to be attacked are military and to cancel or suspend an attack if it becomes apparent that the objective is not military or the attack would be disproportionate. In a fully autonomous system, where there is no human acting as the immediate decision-maker during the engagement sequence, fulfilling the obligations of Article 57 becomes highly problematic. Who is legally executing the precautions in attack when the machine selects the target at machine speed?

Beyond the specific articles of Additional Protocol I, the Martens Clause serves as a crucial legal backstop. Originally appearing in the 1899 Hague Convention and reaffirmed in Additional Protocol I, the Martens Clause dictates that in cases not covered by existing treaties, civilians and combatants remain under the protection and authority of the principles of international law derived from established custom, from the principles of humanity, and from the dictates of public conscience. Proponents of a ban heavily rely on the Martens Clause, arguing that delegating lethal decisions to machines fundamentally violates the "principles of humanity" and deeply offends the "dictates of public conscience."

Furthermore, the Rome Statute of the International Criminal Court provides the framework for prosecuting war crimes. Article 25 of the Rome Statute establishes individual criminal responsibility. If an autonomous weapon commits an act that would constitute a war crime if done by a human, Article 25 requires finding an individual to hold accountable. The difficulty in tracing the chain of causality from a machine's algorithmic decision back to a specific programmer, commander, or political leader highlights the severe inadequacies in current international criminal law regarding artificial intelligence.

Lastly, the Charter of the United Nations itself is relevant. Article 2(4) prohibits the threat or use of force against the territorial integrity or political independence of any state. Article 51 guarantees the inherent right of individual or collective self-defense if an armed attack occurs. The deployment of autonomous swarms, or the unintentional escalation caused by algorithmic interactions (flash war), complicates the determination of when an "armed attack" has occurred and who initiated it, potentially lowering the threshold for the use of force globally.

Important and Major Case Studies

To ground the theoretical debates in practical reality, it is imperative to examine contemporary conflicts where the integration of autonomy and artificial intelligence has transitioned from the laboratory to the battlefield. These case studies illuminate the immediate technical and legal challenges facing the international community.

The Libyan Civil War (2020) provides one of the most heavily scrutinized incidents regarding autonomous weapons. In a report submitted to the United Nations Security Council in 2021, the Panel of Experts on Libya detailed the use of the Kargu-2, a rotary-wing loitering munition manufactured in Turkey. According to the UN report, logistical convoys and retreating forces belonging to the Haftar-affiliated forces were "hunted down and remotely engaged by the unmanned combat aerial vehicles or the lethal autonomous weapons systems such as the STM Kargu-2." The most critical and alarming assertion in the report was that these systems were "programmed to attack targets without requiring data connectivity between the operator and the munition: in effect, a true 'fire, forget and find' capability." While the manufacturer and some analysts disputed whether the weapon operated entirely without human supervision, the UN report marked the first official international documentation suggesting that an autonomous weapon may have been used to hunt and engage human targets independently. This incident immediately elevated the urgency of international regulatory efforts, proving that the technology was proliferating to proxy conflicts.

The Nagorno-Karabakh conflict (2020) between Armenia and Azerbaijan showcased the devastating effectiveness of loitering munitions, heavily blurring the line between human control and autonomy. Azerbaijan extensively deployed systems such as the Israeli-made Harop and various other kamikaze drones. These systems loitered over the battlefield for extended periods, utilizing advanced sensors to detect Armenian air defense systems, armor, and troop concentrations. While many of these strikes involved a human operator confirming the target through a video feed before final impact, the systems performed the vast majority of the search, identification, and tracking functions autonomously. The speed and efficiency with which these semi-autonomous systems dismantled conventional ground forces demonstrated the overwhelming tactical advantage of delegating cognitive tasks to machines, incentivizing militaries worldwide to push for even greater levels of autonomy.

The ongoing conflict in Ukraine has served as an unprecedented testing ground for algorithmic warfare and military artificial intelligence. Both sides have rapidly iterated and deployed thousands of unmanned systems. A notable development is the deployment of drones equipped with artificial intelligence targeting software designed to overcome electronic warfare. As electronic jamming frequently severs the communication link between a drone and its human operator, developers have integrated machine learning models directly onto the drone's onboard processor. Systems like the Saker Scout drone possess software capable of visually identifying specific types of armor or artillery independently. If the drone loses connection to its pilot due to jamming, the onboard algorithm can, theoretically, lock onto a pre-designated target profile and execute the strike autonomously. This operational necessity, driven by the realities of electronic

warfare, represents a direct and pragmatic push toward full autonomy on the battlefield. It demonstrates how combat environments naturally erode the feasibility of maintaining constant, direct human control.

In the context of the Gaza conflict, recent reports have detailed the heavy reliance of the Israeli military on advanced artificial intelligence decision support systems, specifically programs known as "Lavender" and "The Gospel." While the military asserts that these are not autonomous weapons in the strict sense, as a human operator must technically authorize every strike, the systems function by rapidly processing massive amounts of surveillance data, telecommunications, and behavioral patterns to generate target recommendations at a scale and speed impossible for human analysts. Reports suggest that during high-intensity phases, human operators acted primarily as a rubber stamp, approving algorithmic target recommendations with minimal independent verification due to the sheer volume of data. This case study is vital because it highlights the psychological phenomenon of automation bias, where humans inherently trust the output of a sophisticated machine over their own judgment. It demonstrates that even if a human remains technically "in the loop," human control becomes functionally meaningless if the operator cannot critically evaluate the algorithm's targeting logic.

Finally, looking at defensive systems, the continued deployment and evolution of the Aegis Combat System and the Phalanx Close-In Weapon System by naval forces globally remains relevant. These systems operate in environments (open ocean) with very low risk of civilian casualties and are designed to counter targets (incoming supersonic missiles) moving too fast for human interception. They represent the universally accepted "safe" application of autonomous lethal force. However, the regulatory challenge lies in preventing the logic used to justify these strictly defensive, highly constrained systems from creeping into the justification for offensive, unconstrained autonomous weapons operating in complex, civilian-populated urban environments.

Major and Relevant UN Discussions Related to the Agenda

The primary locus of diplomatic activity regarding this agenda has been within the framework of the United Nations Convention on Certain Conventional Weapons (CCW) in Geneva. The CCW is structured to prohibit or restrict the use of specific weapons considered to cause unnecessary suffering or affect civilians indiscriminately.

In 2017, the High Contracting Parties to the CCW established a formal Group of Governmental Experts (GGE) on emerging technologies in the area of Lethal Autonomous Weapons Systems. The mandate of the GGE was to explore and characterize the technological, military, legal, and ethical dimensions of these weapons. For several years, the GGE served as a crucial educational forum, allowing diplomats, military experts, and civil society organizations to debate definitions and share perspectives.

A significant, albeit modest, achievement of the GGE was the adoption in 2019 of the "11 Guiding Principles" regarding emerging technologies in the area of LAWS. These principles

affirmed several critical baseline concepts. They established that International Humanitarian Law continues to apply fully to all weapons systems, including autonomous ones. They affirmed that human responsibility for decisions on the use of weapons systems must be retained, as accountability cannot be transferred to machines. Furthermore, the principles stated that accountability for developing, deploying, and using any emerging weapons system must be ensured in accordance with applicable international law. While these principles were a consensus document, they are non-binding political declarations and lack specific mechanisms for enforcement or clear definitions of what constitutes "human responsibility."

Despite the adoption of the Guiding Principles, the GGE process has largely stalled in recent years. The consensus-based rules of the CCW require unanimous agreement among member states to adopt any binding instrument or mandate. A small but powerful coalition of technologically advanced states has consistently blocked efforts to move from discussion to the negotiation of a legally binding treaty prohibiting or severely restricting autonomous weapons. These states argue that it is premature to ban technology that does not fully exist and reiterate their stance that existing international law is sufficient. Conversely, a growing coalition of states, primarily from the Global South, alongside international organizations, continues to advocate forcefully for a legally binding instrument.

Due to the persistent deadlock within the CCW in Geneva, the diplomatic gravity has slowly begun to shift toward the United Nations General Assembly in New York, which operates on majority voting rather than strict consensus. In December 2023, a historic milestone was achieved when the UN General Assembly adopted Resolution 78/241, titled "Lethal autonomous weapons systems." This was the first ever UNGA resolution dedicated specifically to this issue. The resolution recognized the serious challenges and concerns raised by new technological applications in the military domain and requested the UN Secretary-General to seek the views of Member States and observer States on LAWS and submit a substantive report to the General Assembly. This resolution marks a critical pivot, demonstrating the frustration of the broader international community with the slow pace of the CCW and signaling a desire to elevate the issue to the highest diplomatic levels of the United Nations.

Furthermore, the United Nations Secretary-General, António Guterres, has been a vocal advocate for regulatory action. In his 2023 policy brief, "A New Agenda for Peace," the Secretary-General explicitly called upon member states to conclude a legally binding instrument to prohibit lethal autonomous weapons systems that function without human control or oversight by the year 2026. The Secretary-General has repeatedly stated that machines with the power and discretion to take lives without human involvement are politically unacceptable, morally repugnant, and should be prohibited by international law.

The Human Rights Council has also maintained an active interest in the agenda, focusing heavily on the implications for the right to life and human dignity. Various UN Special Rapporteurs have continued to issue reports warning about algorithmic bias, the lack of accountability, and the dangers of deploying artificial intelligence in law enforcement and military contexts. The convergence of these various UN bodies, the stalled CCW process, the momentum in the General

Assembly, and the explicit calls from the Secretary-General create the immediate context for the deliberations of the Disarmament and International Security Committee.

Questions to Ponder Upon

- How should the international community forge a universal, precise, and technically accurate definition of "Lethal Autonomous Weapons Systems" that effectively distinguishes them from highly automated, yet permissible, defense systems currently in operation?
- At what specific stage in the targeting cycle (surveillance, identification, tracking, selection, engagement) must human control be exercised for it to be considered legally and ethically "meaningful"?
- Given the rapid proliferation of dual-use artificial intelligence research in the commercial sector, what verification mechanisms can a regulatory framework employ to monitor the development phase of military algorithms without stifling legitimate civilian technological innovation?
- Does the deployment of autonomous systems that utilize machine learning, and thus adapt and change their behavior post-deployment, inherently violate the obligation of states to conduct comprehensive legal reviews of new weapons under Article 36 of Additional Protocol I?
- How can the international legal framework address the accountability gap if an autonomous weapon commits an act that constitutes a war crime, considering the impossibility of prosecuting an algorithm and the difficulty of proving malicious intent by a programmer or commander?
- In urban warfare environments where combatants and civilians are intermingled, is it technologically feasible for artificial intelligence to accurately satisfy the principle of distinction, or is this principle inextricably linked to human contextual judgment?
- How should states evaluate the principle of proportionality when utilizing autonomous weapons, given that weighing military necessity against civilian harm requires moral reasoning and value-based judgments that algorithms do not possess?
- Should the international regulatory framework adopt a blanket prohibition on all autonomous lethal systems, or should it adopt a tiered approach, banning certain uses (e.g., targeting humans in urban areas) while regulating others (e.g., anti-armor operations in desolate environments)?
- What specific protocols must be implemented to prevent automation bias, ensuring that human operators acting in a supervisory capacity retain the cognitive ability and authority to override algorithmic targeting recommendations?

- How can a regulatory framework effectively address the risks of adversarial machine learning, where adversaries might subtly manipulate sensor data to cause an autonomous weapon to misidentify targets and violate international humanitarian law?
- What mechanisms are required to mitigate the risk of a flash war, where the interaction of competing autonomous swarms causes a rapid, uncontrollable escalation of conflict before human diplomatic or military intervention is possible?
- Does the inherent opacity of deep neural networks (the black box problem) preclude their use in lethal targeting, given that states must be able to explain and justify the legal basis for every use of lethal force?
- How can the Disarmament and International Security Committee overcome the current diplomatic deadlock caused by the consensus rules of the Convention on Certain Conventional Weapons, and what role should the General Assembly play in forcing progress?
- If a legally binding treaty is deemed politically impossible in the short term, what specific soft law instruments, codes of conduct, or technical standards can be immediately implemented to constrain state behavior and reduce risk?
- How should international regulations address the risk of non-state actors, insurgencies, or terrorist organizations acquiring and deploying crude but lethal autonomous weapons systems constructed from commercially available drone and software technology?
- To what extent does the Martens Clause, which protects civilians based on the principles of humanity and the dictates of public conscience, render the deployment of fully autonomous weapons inherently illegal under customary international law?
- If an algorithm is trained on data that contains implicit historical or demographic biases, resulting in the weapon disproportionately targeting specific ethnic groups or populations, how does international law address this systemic violation of human rights and the laws of war?
- In the event that a communication link is severed by electronic warfare, should autonomous weapons be legally required to possess a fail-safe mechanism that forces them to abort the mission and return to base, rather than autonomously engaging pre-selected target profiles?